

Enhanced Network Intrusion Detection Using a Hybrid CNN-LSTM Approach on the UNSW-NB15 Dataset

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Abstract—In the era of rapid technological advancements, data and information have become significant assets that require continuous protection from unauthorized access and malicious attacks. Network Intrusion Detection Systems (NIDS) can effectively manage the increasing data demands assuring robust security. However, traditional or manual methods are often inadequate to cope with the sophisticated and evolving nature of cyber threats due to their inability to quickly process large volumes of data. Traditional machine learning models are widely used for this purpose. Recently deep learning approaches have become popular in classification tasks showing promising results. This paper introduces a hybrid deep-learning approach for detecting various network intrusions using binary and multi-class (ten classes) classification on the well-known UNSW-NB15 dataset. The proposed hybrid CNN-LSTM model demonstrates significant improvements in intrusion detection, achieving an accuracy of 97.19% for binary classification and 87.70% for multi-class classification, particularly in handling complex attack patterns and reducing false positives. Evaluated using metrics such as precision, recall and F1-score, the model consistently performs robustly across nine distinct attack categories. These results highlight the model's effectiveness in establishing enhanced network security.

Index Terms—network intrusion detection, hybrid model, anomaly detection, deep neural network, CNN-LSTM, UNSW NB15

I. INTRODUCTION

The Internet has long been an essential component of modern life, playing a critical role in communication, commerce and technological advancements. In nearly every aspect of life, people rely on it for online transactions, payments, cloud computing, data transfer and more. However, the alarming evolution of cyber crimes has opened the potential of Network Security.

Anomaly attacks are handled and prevented with much protection utilizing the Intrusion Detection System. Among the four types of IDS, the Network Intrusion Detection System considers network-based malicious attack analysis to prevent the system from any unauthorized access. The rapidly expanding complex challenges and cyber-attack complications constantly reduce the effectiveness of previously built traditional anomaly detection systems. As these attacks are growing

stronger each day, those NIDS techniques require continuous improvement and testing to build a stronger detection system that will protect all the systems, detecting and preventing any malicious attack.

Traditional prevention techniques such as firewalls, encryption methods and anti-viruses are signature-based and are insufficient to tackle all types of cyber attacks struggling with identifying novel and evolving threats stating the need for advanced detection systems. With this purpose, deep learning-based systems are encouraged to structure an NIDS considering their advanced performance and higher accuracy rates. The introduction of deep learning shown in figure 1 has revolutionized the field of intrusion detection as these models can automatically learn and extract complex patterns from huge datasets.



Fig. 1: Intrusion Detection Systems

The main objective of choosing these models is to enhance the detection rate by efficiently recognizing temporal and spatial patterns in network traffic data, improving both binary and multi-class classification accuracy. This paper develops a hybrid CNN-LSTM model for network intrusion detection using the UNSW-NB15 dataset. This combines two of the very well-known algorithms of CNN and LSTM. The system is implemented to achieve a superior performance with much higher accuracy and lower false positive rates.

The proposed model can handle complex challenges and detect attacks due to their special ability to adapt learning by itself with an incredible self features extraction ability.

II. RELATED WORKS

Intrusion detection is a prevalent topic that remains proactive in modernizing its techniques due to its importance in handling and protecting user data from various sources. Some

existing works suggest different model implementations and performance evaluations generalizing an idea in this domain.

A CNN architecture-based model is implemented by Xiao et al, [1] enhancing detection precision and decreasing classification duration, leading to the development of real-time intrusion detection. This study utilized the KDD-CUP99 Dataset. Seo et al, [2] presented a real-time Network Intrusion Prevention System (NIPS) based on hybrid machine learning techniques. Their approach combined multiple machine learning algorithms to detect and prevent network intrusions using two comprehensive datasets, namely, the UNSW-NB15 dataset and the CICIDS2017 dataset. Another study, Vinayakumar et al, [3] implied the DNN model focusing on the advantages of these sequential deep learning models. Adding up the layers improved the overall performance though an absence of fine-tuning model parameters and feature selection and a lack of detailed insights into malware characteristics is visible. The research study [4] proposed a CNN-LSTM hybrid approach for detecting Denial of Service (DoS) attacks in Wireless Sensor Networks (WSNs) focusing only on the DoS types of attack making the approached model unacquainted with the various kinds of real-time attacks and cyber threats.

These studies require enhanced adaptability in diverse network environments to manage various and expansive networks. Furthermore, the difficulty in interpreting and understanding model decisions is still a challenge in maintaining trust and transparency in NIDS operations.

As the significance of this domain emphasizes constructing neural networks that project the similarity of the human brain, deep learning-based algorithms are now mostly popular and further implemented for remarkable performance. Also, these studies stated some limitations as they face difficulties handling huge amounts of records establishing scalability and robustness. They predominantly focus on specific types of attacks or datasets. Most of these papers work with similar datasets on different proposed models. Rectifying these areas of research will aid in creating intrusion detection systems that are better equipped to handle new cyber threats in real-time settings.

This study solely focuses on improving these research gaps in existing cases to construct an efficient deep learning model for achieving superior performance metrics, including higher accuracy and lower false positive rates.

III. METHODOLOGY

A. Proposed Approach

The proposed methodology is shown in figure 2 below and discussed in detail in the upcoming sections. Based on this methodology the model is implemented to identify two different classes: Attack and Non-Attack. For multiclass classification, the non-attack class is divided into individual attack types.

The methodology includes preprocessing for data handling, followed by training and evaluation to identify these anomalies. To further enhance classification accuracy, sophisticated

techniques are integrated to capture intricate data patterns resulting in improved model precision and robustness.

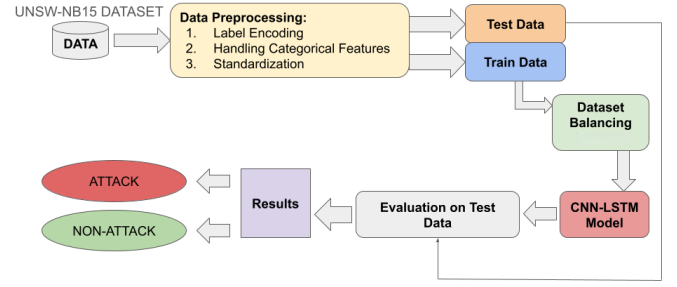


Fig. 2: Proposed Methodology

Figure 3 illustrates the model composition and decision-making process highlighting the classification of different classes.

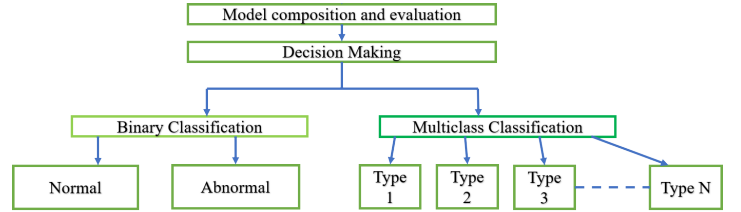


Fig. 3: Model Evaluation and Decision Making

The main purpose of this research is to efficiently identify all those attacks to protect the network from vulnerabilities and anomalies ensuring network security.

B. Dataset Description

Numerous datasets are available in the domain of network intrusion detection, such as the KDD Cup 1999 Dataset, NSL-KDD Dataset, UNSW-NB15 Dataset, CIC-IDS2017, etc. Considering the factors of data availability, relevance to the problem domain and data quality, the UNSW-NB15 Dataset [5] is chosen for our model to form the foundation of our model development and evaluation.

The UNSW-NB15 dataset is more recent and robust in identifying network intrusions. The below Table I contains the detailed information [5] about the dataset:

TABLE I: UNSW-NB15 Dataset Specifications

Dataset Name	UNSW-NB15
Dataset Type	network intrusion dataset
Training Set Size	175,341 records
Testing Set Size	82,332 records
Total Number of Records	257,673 records
Feature Number	42 features

This data collection contains a combination of real-world modern Normal and Abnormal Attack activities that existed

in the network traffic. It comprises actual statistical, content-based and time-based attributes related to network packets, such as flow duration, source and destination IP addresses, protocols and more, extracted from a large-scale academic network environment containing nine attack types, name, Denial of Service (DoS), Fuzzers, Analysis, Backdoors, Exploits, Generic, Reconnaissance, Shellcode and Worms.

C. Class Distribution

The UNSW-NB15 dataset is preprocessed for two types of classification: Binary Classification and Multi-class Classification. For both types, two different types of approaches are taken.

Binary Class deals with two different classes, Normal and Abnormal whereas the Multi Class deals with ten different classes. The dataset distribution for both train and test types is given below in figure 4a and 4b for binary classification respectively.

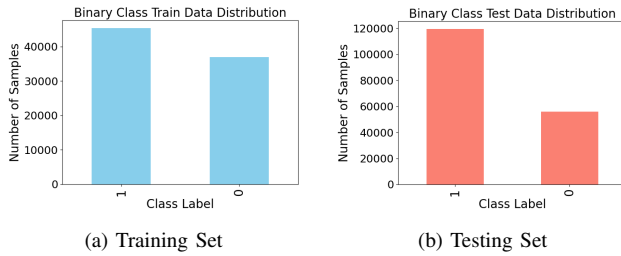


Fig. 4: Binary Class Distribution

Figure 5a and 5b show the dataset distribution for both training and testing sets in multi-class classification respectively.

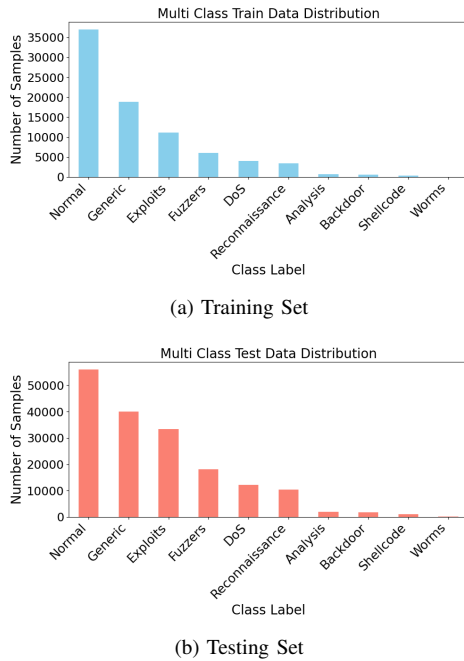


Fig. 5: Multi Class Distribution

The above figures illustrate that the dataset is quite imbalanced, showing unequal distributions among its classes. This can greatly impact achieving good accuracy rates which is why the imbalanced data is handled while training using class weights. It mainly improves by assigning higher weights to the minority classes and lower weights to the majority classes during the training process, highly increasing the model's performance.

D. Dataset Preprocessing

The UNSW-NB15 dataset was preprocessed involving some steps before the model implementation process. As the dataset contains both numerical and categorical type data, the categorical records are converted into numerical form using Label Encoder. Also, the 'id' and 'label' columns are dropped from the dataset as they are irrelevant for modeling. Features with categorical values are converted into numerical representations using one-hot encoding. Thus the categorical features are handled. All numerical features are standardized using StandardScaler to have zero mean and unit variance which helps accelerate the convergence of the training algorithms.

These steps contribute to attaining high accuracy values to form balanced and well-furnished data. It facilitates the model implementation steps ensuring the robustness of the model.

E. Train Test Splitting

Train Test Splitting is a crucial step that takes some parts of the train data as validation data to check whether the model is overfitting. This step allows evaluation of the model during the training phase so that any misclassification or bad performance is identified and fixed during the training session. The UNSW-NB15 dataset is available online and a partition from this dataset was separated as a training set and testing set, namely, UNSW_NB15_training-set.csv and UNSW_NB15_testing-set.csv respectively. For this study, both files are used separately to train and evaluate the model performance. The paper used 20% of the training data as the validation data and separately used the unseen data for testing.

F. Model Description

The proposed hybrid CNN-LSTM model consists of convolutional layers, followed by LSTM layers to capture temporal dependencies. The model combines the strengths of CNNs in feature extraction with the sequential processing capabilities of Long Short-Term Memory (LSTM) networks. The model is trained using binary cross-entropy loss for binary classification and categorical cross-entropy loss for multi-class classification. The structure of the model shown in figure 6 implemented two models, combining both the Convolutional Neural Network (CNN) and Long Short Term Memory (LSTM) components.

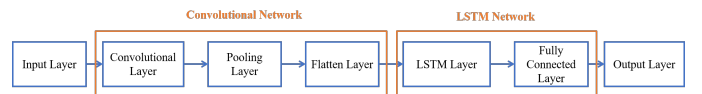


Fig. 6: Proposed CNN-LSTM Model

CNNs, also known as Convolutional Neural Networks, focus on analyzing multi-dimensional array information and are frequently used in applications like image recognition and natural language processing. This deep learning algorithm excels in discovering connections within local features, which ultimately improves the precision of feature extraction. A convolutional neural network is made up of a starting layer that receives input, multiple intermediate hidden layers and an output layer.

On the other hand, LSTMs are a type of RNNs specifically created to understand and retain important information over an extended period in sequential data which makes them appropriate for language modeling and analyzing time series data. They incorporate unique memory cells and gates that control information flow, allowing them to store data for long periods.

By integrating CNN and LSTM layers, the model effectively learns representations of complex temporal patterns in network traffic data, leading to improved detection performance.

G. Model Architecture

The architecture of the proposed CNN-LSTM network model is a combination of convolutional and recurrent layers. The input layer processes the 40 features, each reshaped to (40, 1) for compatibility with the Conv1D layer. The convolutional layers (Conv1D) capture spatial features from the input data, followed by LSTM layers that learn temporal dependencies.

Batch normalization is applied to each convolutional and LSTM layer to improve training stability and accelerate convergence. By randomly dropping 50% of the input units during the training process, the Dropout layers (with a dropout rate of 0.5) prevent overfitting. The activation function used in the Conv1D and dense layers is ReLU, while the LSTM layers use the tanh activation function. The final output layer uses the softmax activation function to classify the data into ten different classes. The model architecture is shown in Table II below.

TABLE II: Proposed CNN-LSTM Model Summary

Layer (type)	Output Shape	Param #
conv1d_2 (Conv1D)	(None, 40, 128)	512
batch_normalization_2 (BatchNormalization)	(None, 40, 128)	512
dropout_2 (Dropout)	(None, 40, 128)	0
lstm (LSTM)	(None, 40, 128)	131,584
batch_normalization_3 (BatchNormalization)	(None, 40, 128)	512
dropout_3 (Dropout)	(None, 40, 128)	0
flatten_1 (Flatten)	(None, 5120)	0
dense_6 (Dense)	(None, 256)	1,310,976
batch_normalization_4 (BatchNormalization)	(None, 256)	1,024
dropout_4 (Dropout)	(None, 256)	0
dense_7 (Dense)	(None, 128)	32,896
batch_normalization_5 (BatchNormalization)	(None, 128)	512
dropout_5 (Dropout)	(None, 128)	0
dense_8 (Dense)	(None, 10)	1,290

From the model summary above, a better understanding of each layer is obtained along with the knowledge of a

total of 1,373,194 parameters among which the number of Trainable and Non-trainable parameters are 1,372,170 and 1,024 respectively.

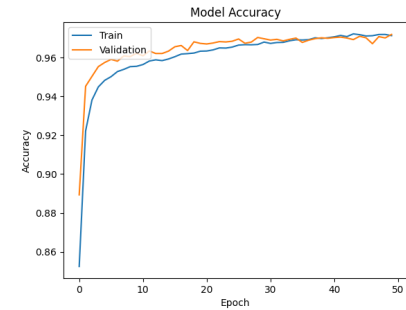
H. Training and Evaluation

In this segment, the model is trained to prepare it for the unseen data. This paper performed 50 epochs for the CNN-LSTM model and trained with the training dataset. The model was trained by adjusting its parameters to minimize the loss function, using Early Stopping to avoid overfitting. Class weights were applied to handle imbalanced classes during training. Additionally, a learning rate of 0.001 was used to ensure gradual and stable updates to the model's parameters. The accuracy after each epoch is observed which justifies the overall model performance of how well it is conducted during the training period. These measures justify that the proposed model is performing well both for training and testing data.

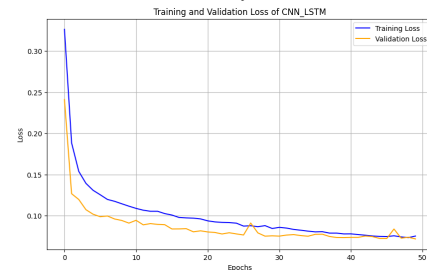
IV. EXPERIMENTAL RESULTS

This section illustrates the performance evaluation where the custom-built CNN-LSTM model achieved an accuracy of 97.17% for Training and 97.19% for Validation data, respectively for binary class classification.

The model was effectively run on the preprocessed dataset for 50 epochs to achieve higher accuracy. This model also incorporates early stopping to ensure optimal performance. The accuracy and loss curves for both binary and multi-class classification are displayed below in figure 7 and figure 8 respectively, indicating that the validation loss is lower than the training loss for this model. The training loss reflects how well the model fits the training data.



(a) Accuracy Curve



(b) Loss Curve

Fig. 7: Binary Class Accuracy and Loss Curve of CNN-LSTM Model

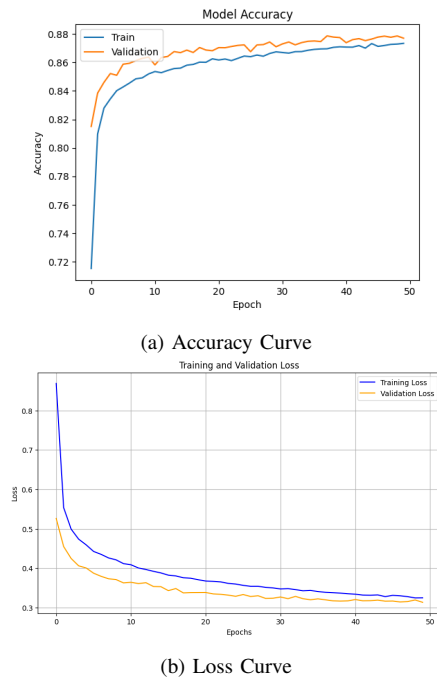


Fig. 8: Multi Class Accuracy and Loss Curve of CNN-LSTM Model

The model, with its combined layers, is likely to achieve higher accuracy than many other existing models. A decrease in training loss indicates reduced error between predictions and true labels. When evaluated using the testing set of the UNSW-NB15 dataset for binary classification, the trained model attained a test accuracy of 88.91%, with precision, recall and f1-score of 99.29%, 84.31% and 91.19% respectively.

For multi-class classification, the attained accuracy for training and validation data was 87.42% and 87.70% respectively. Using the testing set of the UNSW-NB15 dataset, the trained model is evaluated, and the test accuracy, precision, recall and f1-score are 75.16%, 76.56%, 75.16% and 72.50% respectively.

Another evaluation approach is the Confusion Matrix which is used for understanding the performance of a classification algorithm by providing a summary of both correct and incorrect classifications. The training and testing set performance in binary class classification is shown in figure 9.

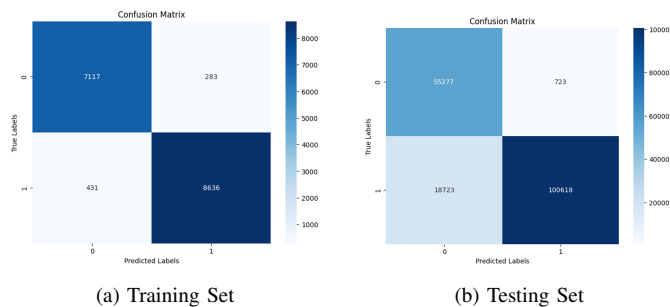


Fig. 9: Confusion Matrix of Training and Testing Set in CNN-LSTM Model for Binary Class Classification

The matrix is split into four sections: True Positives, True Negatives, False Positives and False Negatives. For multi-class classification, the training and testing set performance of the confusion matrix is shown in figure 10 and 11.

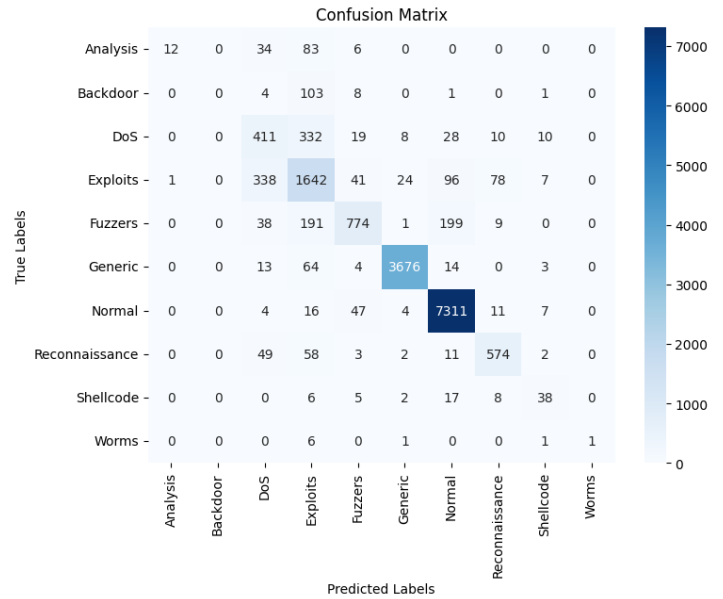


Fig. 10: Confusion Matrix of Training Set in CNN-LSTM Model for Multi-Class Classification

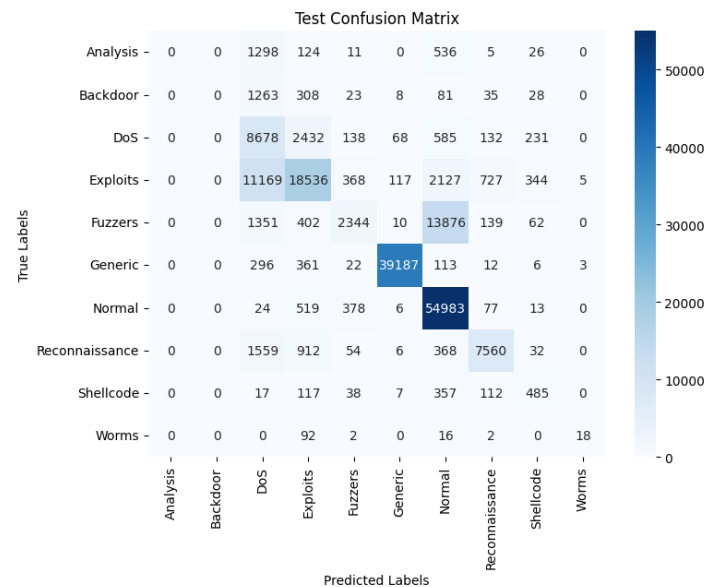


Fig. 11: Confusion Matrix of Testing Set in CNN-LSTM Model for Multi-Class Classification

The ROC curve (Receiver Operating Characteristic Curve) in figure 12 illustrates the case for binary classification, achieving an AUC Score of 0.98. In the case of multi-class classification, the model attained an AUC Score of 0.94, confirming its excellent performance.

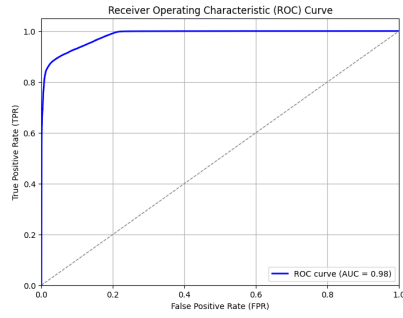


Fig. 12: ROC Curve of CNN-LSTM Model

V. DISCUSSION AND RESULT ANALYSIS

This section analyzed the performance matrices comparing each model and finding the test accuracy, recall, precision and f1-score, key factors for assessing the classification performance based on how well the model performs on unseen testing set data.

TABLE III: Test Data Performance Comparison of Proposed CNN-LSTM Model for Binary and Multi-class Classification

Class	Train Acc	Val Acc	Test Acc	Recall	Precision	F1-Score
Binary	97.17%	97.19%	88.91%	84.31%	99.29%	91.19%
Multi	87.42%	87.70%	75.16%	75.16%	76.56%	72.50%

Based on the above observation, it is evident that the proposed CNN-LSTM model excelled in achieving efficient performance for both binary and multi-class anomaly detection. This accomplishment underscores the model's capability to effectively detect anomalies across various scenarios. This study emphasizes the increasing popularity of hybrid models in recent days. The result analysis shows that hybrid models outperform individual models.

The difference between the training set and testing set accuracy is because of better performance in the case of known data. When unseen data is provided, the accuracy is reduced slightly. The comparison of existing models used in previous research is presented in Tables IV and V for Binary and Multi-class classification respectively.

TABLE IV: Performance comparison of proposed techniques with related works for Binary Classification

Author	Dataset	Algorithm	Val Acc	Test Acc
Al-Turaiki et al. [6]	UNSW-NB15	CNN	-	89.26%
Merna et al. [7]	UNSW-NB15	CNN	-	91%
Du et al. [8]	UNSW-NB15	CNN-LSTM	94%	-
This paper	UNSW-NB15	CNN-LSTM	97.19%	88.91%

TABLE V: Performance comparison of proposed techniques with related works for Multi-Class Classification

Author	Dataset	Algorithm	Val Acc	Test Acc
Wu et al. [9]	KDDTest+	CNN	79.48%	-
Wu et al. [9]	KDDTest-21	CNN	60.71%	-
Al-Turaiki et al. [6]	UNSW-NB15	CNN	-	68.25%
Kumar et al. [10]	UNSW-NB15	Integrated Model	-	84.83%
Du et al. [8]	UNSW-NB15	CNN-LSTM	82.9%	-
This paper	UNSW-NB15	CNN-LSTM	87.70%	75.16%

From the above comparison Tables IV and V, the proposed CNN-LSTM architecture has shown the benchmark performance acquiring very high accuracy and performance rates. It serves as an example to project the concept of detecting attacks detection using deep learning techniques manifesting the advancement of recent technology and its need in data protection.

VI. CONCLUSION

In summary, the custom-built CNN-LSTM model has shown a significant impact in achieving high accuracy for both binary and multi-class classification. The overall accuracy for binary classification is 88.91% and for multi-class classification is 75.16%, outperforming existing models. These results highlight the model's effectiveness in distinguishing between normal and malicious network traffic, as well as accurately classifying various types of attacks. The success of this approach underscores its potential for enhancing network security. These deep learning methods, when combined into a hybrid approach, perform well due to their advanced detection and feature engineering capabilities, offering significant advantages in network intrusion detection systems. Future work will explore the integration of additional features and the application of transfer learning to further enhance intrusion detection systems.

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